

Luyao Zhang

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Research Interests

Planet formation; gravitational instability and disc fragmentation; binary and multiple-star formation; planet–disc interactions; planet migration and scattering; tidal disruption; free-floating planets; FU Ori accretion outbursts; stellar dynamics and *Gaia* astrometry.

Education

PhD in Physics (Astronomy)

Sep 2024 – Present

University of Leicester, School of Physics and Astronomy, UK

Supervisor: Prof. Sergei Nayakshin

BSc in Astronomy

Sep 2020 – Jun 2024

Shanghai Jiao Tong University (SJTU), Shanghai, China

Supervisor: Prof. Fabo Feng

Peer-Reviewed Publications

3.1 First-author

1. **Zhang, L.**, Nayakshin, S., Baruteau, C., Thebault, P., & Vorobyov, E. I. (2026).

Disc fragmentation – III. The need for a new paradigm for formation of planets within close binary systems

MNRAS, **550**(4), stag1363. doi:10.1093/mnras/stag1363

2. **Zhang, L.**, Feng, F., Rui, Y., Xiao, G.-Y., & Wang, W. (2025).

A Cartesian Catalog of 30 Million Gaia Sources Based on Second-order and Monte Carlo Error Propagation.

RAA, **25**, 055010. doi:10.1088/1674-4527/adc6da

3.2 Other contributions

1. **Nayakshin, S.**, **Zhang, L.**, Calovic, A., Lee, H., Baruteau, C., Meru, F., & Mayer, L. (2026).

Disc fragmentation – II. Ejection of low-mass free-floating planets from growing binary systems.

MNRAS, **546**(2), stag043. doi:10.1093/mnras/stag043

2. **Guo, Y.**, **Zhang, L.**, Feng, F., Li, Z.-Y., Pomazan, A., & Yang, X. (2025).

Search for Past Stellar Encounters and the Origin of 3I/ATLAS.

AJ, **170**, 362. doi:10.3847/1538-3881/ae1833

Research Projects and Work in Progress

Formation and Survival of S-type Planets surround secondary in Growing Binary Systems *In preparation*

- Study the formation of S-type planets in close binaries and determine how planet parameters affect planetary survival.

FU Ori Outbursts in Growing Binary Systems

In preparation

- Investigate mass accretion and FU Ori-like outbursts produced during the formation and dynamical evolution of young binary systems.

- Identify and characterise candidate stellar encounters using *Gaia* DR3 astrometry and radial velocities.

Selected Talks and Conference Presentations

UKI Disc Meeting 2026 , Cambridge, UK. [Poster]	<i>Jul 2026</i>
PLATO 2025 Meeting , Queen Mary University of London, UK Talk: <i>A Robust Formation Channel for Planets in Binary Systems</i>	<i>Jan 2026</i>
EPF 2025 Meeting , Shanghai, China. [Poster]	<i>Dec 2025</i>
COST Action PLANETS WG2 Code Introduction Meeting , Online Online presentation: <i>Dusty FARGO-ADSG: A 2D Hydrodynamical Code with Dust</i>	<i>Oct 2025</i>
UKI Disc Meeting 2025 , Hertford, UK. [Poster]	<i>Sep 2025</i>
Multiplicity in Young Stars Conference , Copenhagen, Denmark. [Online participant]	<i>Aug 2025</i>
Midlands Disc Meeting , University of Warwick, UK Talk: <i>Tracking the Evolution of Gas Clumps in Gravitationally Unstable Discs</i>	<i>May 2025</i>
STFC Introductory Summer School in Astronomy , Keele University, UK. [Selected participant]	<i>Sep 2024</i>

Computational and Technical Skills

Programming	Python, C, C++, R, Bash
Hydrodynamics	FARGO-ADSG
Orbital dynamics	galpy, REBOUND
Astronomical data	<i>Gaia</i> DR3, catalogue cross-matching, stellar kinematics, astrometric uncertainty propagation

Languages

Mandarin Chinese:	Native
English:	Professional working proficiency